



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

TCP/IP PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Networking

TOTAL: 10 QUESTIONS

Q1 What is TCP/IP and what problem in Networking does it solve?

Q6 How does TCP/IP handle reliability and high availability?

Q2 What are the core components or concepts in TCP/IP?

Q7 What observability does TCP/IP provide out of the box?

Q3 How does TCP/IP compare to its main alternatives?

Q8 What are common performance bottlenecks in TCP/IP?

Q4 How do you get started with TCP/IP for a new project?

Q9 What security best practices apply when running TCP/IP?

Q5 What configuration options most affect TCP/IP's behavior in production?

Q10 What mistakes do teams commonly make when adopting TCP/IP?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 TCP/IP is a tool or platform that

Mitigation

TCP/IP is a tool or platform that automates or simplifies a specific concern in the Networking domain, reducing manual effort and operational complexity for engineering teams.

A6 TCP/IP provides HA through leader election, replication, or stateless horizontal

availability

TCP/IP provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 TCP/IP has key building blocks that work

Primitives

TCP/IP has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 TCP/IP exposes metrics (Prometheus endpoint or built-in dashboard), structured

information

TCP/IP exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 TCP/IP trades off simplicity against feature richness

Hardening

TCP/IP trades off simplicity against feature richness differently than its alternatives. Choose TCP/IP when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection

pooling

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install TCP/IP via its package manager or

Gotchas

Install TCP/IP via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run TCP/IP with the principle of least

Assessment

Run TCP/IP with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values,

configuration

retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial

and error

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.